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Autoencoder with Clustering on Image Deblurring/Denoising

Convolutional Autoencoder with latent space clustering. Aiming to remove heterogeneous blur/noise in images.

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Motivation

- Restoring an image from blurring/noising that is not globally exerted.

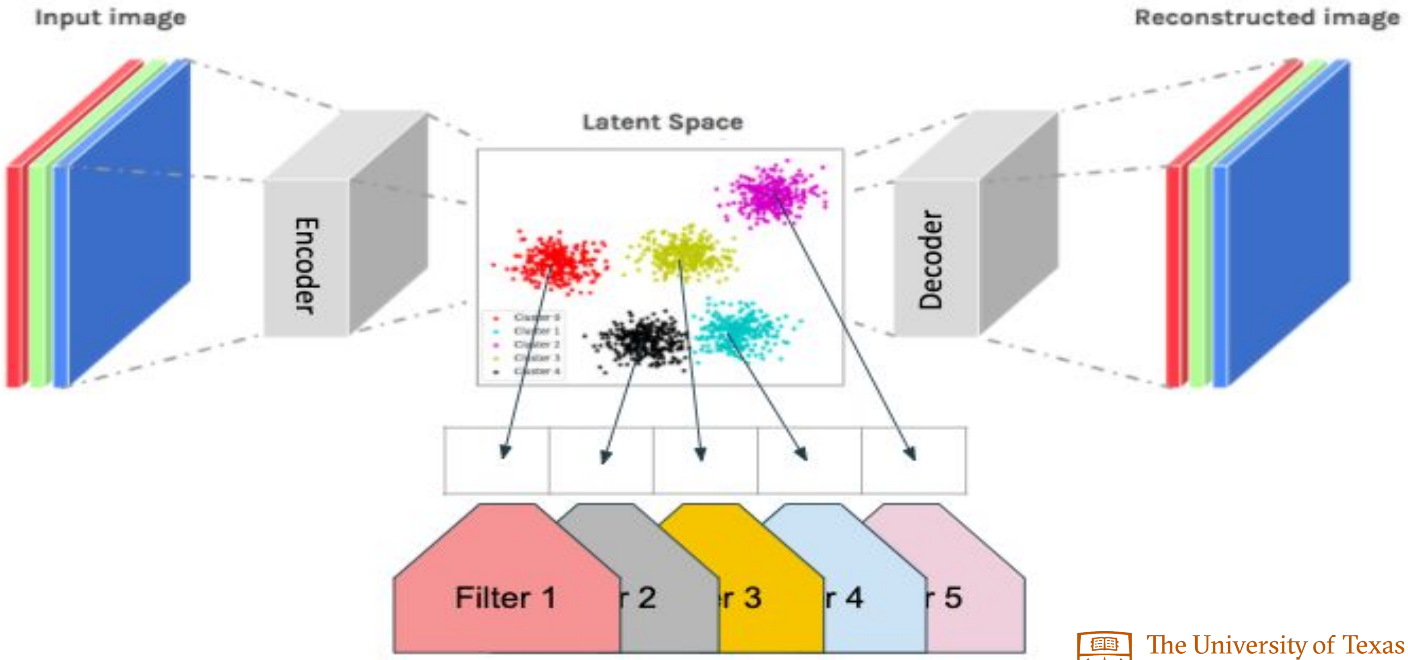
Heterogeneous Noised/Blurred Image

Motivation

- Restoring an image from blurring/noising that is not globally exerted.
- Block-wise (or pixel-wise) manipulation.
- Improving the quality of reconstruction and reducing runtime (no need to apply deblurring filter to all the pixels).

Block-wise Noised/Blurred Image

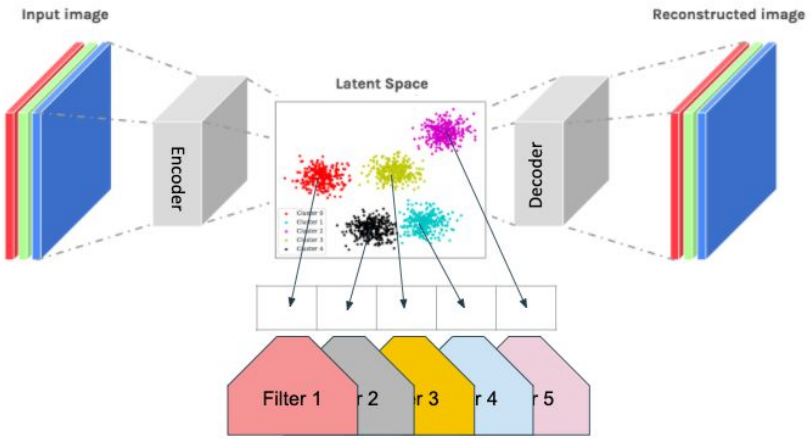
Network Structure



Multi-filter Autoencoder Architecture

Training Scheme

- The encoder and the decoder is trained to generate clusters in the latent space.
- The input data are classified into clusters.



Multi-filter Autoencoder Architecture

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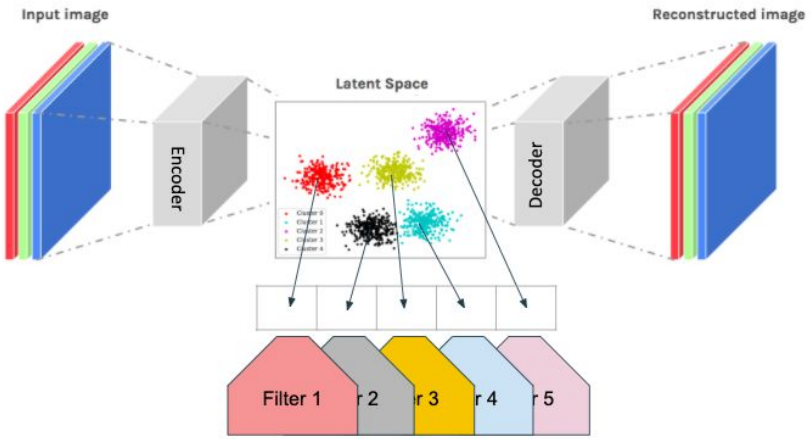
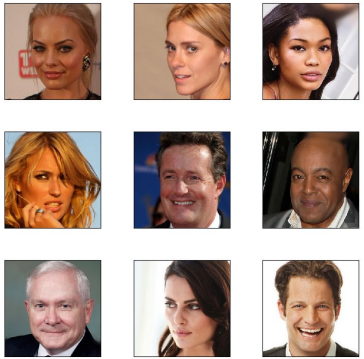


Multi-filter Autoencoder Architecture

- The encoder and the decoder is trained to generate clusters in the latent space.
- The input data are classified into clusters.
- Each filter is trained only using the data from the cluster corresponding to this filter.
- Clusters are identified respectively to determine the exact distributional shift to apply. Thus, heterogeneous blurring can be resolved.

Experiment

CelebFaces Attributes Dataset (CelebA) in 256 x 256 Resolution.



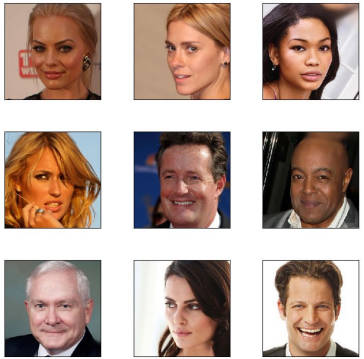
CelebA

Experiment

CelebFaces Attributes Dataset (CelebA) in 256 x 256 Resolution.

Mean Square Error (MSE) for Reconstruction Loss and Soft Normalized Cut Loss for Clustering Loss.

$$Ncut_K(V) = \sum_{k=1}^{K} \frac{cut(A_k, V - A_k)}{assoc(A_k, V)}$$



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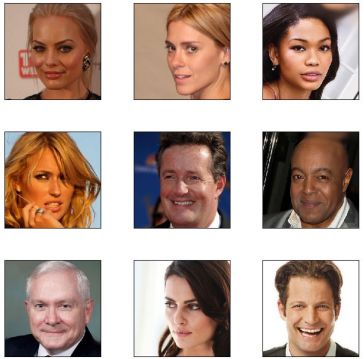
Quality Metrics:

- PSNR (Peak Signal-to-Noise Ratio)
- SSIM (Structural Similarity)
- UQI (Universal Quality Index)

$$PSNR = 10 \cdot \log_{10}\left(\frac{MAX_I^2}{MSE}\right)$$

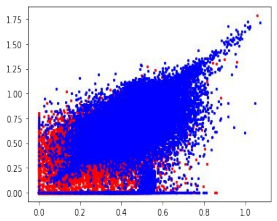
$$\mathrm{SSIM}(x, y) = \frac{(2\mu_x\mu_y + c_1)(2\sigma_{xy} + c_2)}{(\mu_x^2 + \mu_y^2 + c_1)(\sigma_x^2 + \sigma_y^2 + c_2)}$$

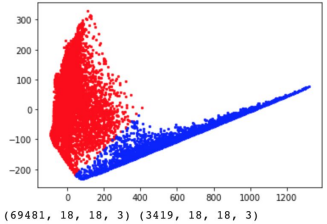


CelebA

Result

Reconstruction Result





Clustering Result

Quality Metric	Noised Image	Traditional Autoencoder	Multi-Filter	Percentage
PSNR	11.37	25.51	26.19	92%
SSIM	0.5593	0.7337	0.7613	98.67%
UQI	0.5965	0.9555	0.9642	100%

Discussion

Conclusion:

The results indicate that the multi-filter model does achieve better performance than the single-filter autoencoder.

Future Studies:

Increase the number of clusters and filters for images with different degrees of blurring.

Reduce runtime by utilizing filters with different complexity.

Acknowledgement

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Reference

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[2] Ding, F., Luo, F. (2019). Clustering by Directly Disentangling Latent Space. arXiv preprint arXiv: 1911.05210. https://arxiv.org/abs/1911.05210.

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